Electric and hybrid cars

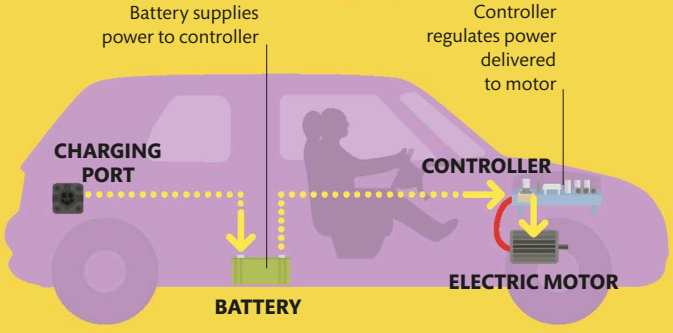
Most cars are powered by an internal combustion engine burning gasoline or diesel fuel. However, concerns about the harmful air pollution produced by these engines has led to the development of less-polluting electric and hybrid cars.

WHEN WAS THE FIRST HYBRID CAR BUILT?

Engineer Ferdinand Porsche built the world's first hybrid car in 1900. He named it the Lohner-Porsche Semper Vivus ("always alive").

Electric cars

An electric car is powered by one or more electric motors. The motor is connected to a rechargeable battery pack. Electric cars are simpler than conventional piston-engine cars because they do not need a fuel system, ignition system, water cooling system, or oil lubrication system. A transmission is not necessary either as, unlike internal combustion engines, electric motors deliver the maximum turning force (torque) across their whole speed range.



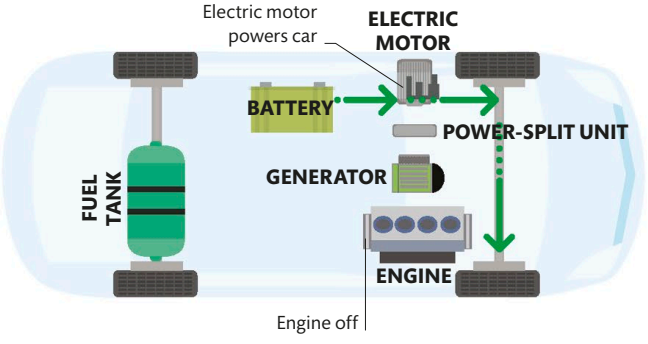
Hybrid cars

A hybrid car has two or more different power sources driving the wheels—an internal combustion engine and at least one electric motor. There are two main types of hybrid cars. A series hybrid is always powered by its electric motor. The role of its internal combustion engine is to run a generator that produces electricity to power the electric motor and charge its battery. The second type of hybrid car, the more popular parallel hybrid (see right), can be powered by either one of its power sources, or both can be used together when maximum power or acceleration is needed.

KEY

- Electric power
- Power from internal combustion engine

Pulling away
Most hybrid cars start moving by using only their battery-powered electric motor. The internal combustion engine is not needed. For short trips at low speeds, electric power alone may be used for the whole trip.



Accelerating
If rapid acceleration is needed, the internal combustion engine starts. The car's wheels are driven by the power of both the engine and the electric motor. The engine also runs a generator that recharges the electric motor's battery.

